SEP Spike Gate Optuna Optimisation

Technical reference for using Spike Gate profile evidence to optimise global SEP foreground masking.

Purpose: This project repeats the Spike Gate plus Optuna optimisation idea, but uses SEP/SExtractor-style segmentation instead of MTObjects. The result is a resumable optimiser that chooses SEP parameters based on spike coverage and data-loss penalties.

Code and Outputs

Item	Path or role
Optimiser script	Foreground Masking/optimise_sep_spike_gate_parameters.py
SEP implementation	Foreground Masking/interactive_sep_spike_gate_parameter_tester.py provides SEP masks and Spike Gate helpers.
Default output folder	Remove foreground objects/sep spike optimisation/<timestamp>
Best parameter file	sep_spike_optimisation_best.json
Study database	sep_spike_optimisation_study.sqlite3, reused by --resume-output-dir

Workflow

1. Load usable galaxies from the S4G manifest, or the explicit list supplied with --names.
2. Build a deprojected, bar-aligned profile and use Spike Gate to mark narrow positive spike samples.
3. For each Optuna trial, run global SEP segmentation with the sampled parameter set.
4. Project the two-dimensional SEP mask back onto the bar-major profile.
5. Score the trial by rewarding spike coverage and penalising non-spike profile masking, image masking, and profile change.
6. Write per-trial summaries, per-galaxy details, the best JSON, and a resumable Optuna SQLite study.

Spike Gate Criteria

Criterion	Default	Meaning
Finite positive sample	Required	Only valid positive profile samples are eligible.
Central exclusion	8 arcsec	Protects central galaxy light from becoming optimisation target evidence.
Local peak	Candidate >= immediate neighbours	Requires a narrow local maximum in the bar-major profile.
Neighbour excess	25 percent above 4-15 arcsec neighbourhood	Requires contrast against the surrounding radial profile.
Side-drop check	40 percent above +/-3 samples	Rejects broad shoulders and slowly varying galaxy structure.
Window expansion	+/-2 samples	Scores a small radial region around each detected spike.

SEP Process

- The detection image is either the original image or a smooth-model residual, controlled by --detect-on.
- SEP estimates a background with the selected background box size, subtracts it, and extracts connected sources above detect_thresh.
- A Gaussian-like filter kernel of filter_size is applied during extraction.
- SEP deblends sources using deblend_nthresh and deblend_cont.
- Detected segments are filtered by max_area, max_elongation, and exclude_center_pixels.
- The final mask is dilated by dilation_radius and then scored against the Spike Gate profile target.

SEP parameter	Optimised range	Effect
detect_thresh	0.5 to 8.0 sigma	Lower values catch fainter sources but can over-mask galaxy structure.
minarea	1 to 80 pixels	Minimum connected area SEP requires for a source.

SEP parameter	Optimised range	Effect
deblend_nthresh	8 to 64	Number of deblend thresholds.
deblend_cont	0.0001 to 0.1, log sampled	Controls how readily SEP separates blended sources.
back_size	16 to 256 pixels	Background mesh size used by SEP.
filter_size	1, 3, 5, 7, or 9 pixels	Smoothing kernel size for SEP extraction.
dilation_radius	0 to 8 pixels	Expands retained segments to cover object wings.
max_area	20 to 5000 pixels	Rejects broad detections likely to be galaxy structure.
max_elongation	1.5 to 20.0	Rejects highly elongated detections.

Objective Function

Objective: 12 x (1 - mean_spike_coverage) + 4 x mean_non_spike_profile_fraction + 2 x mean_masked_fraction + 1.5 x mean_profile_affected_fraction + 1 x mean_profile_change

Metric	Interpretation
mean_spike_coverage	Fraction of Spike Gate profile samples covered by the SEP mask. Higher is better.
mean_non_spike_profile_fraction	Fraction of non-spike profile samples masked. Lower protects the science profile.
mean_masked_fraction	Fraction of all image pixels masked. Lower means less global data loss.
mean_profile_affected_fraction	Fraction of the bar-major profile touched by the mask.
mean_profile_change	Median log-intensity change after log-linear bridging across masked samples.

Running the Optimiser

Initial 20-Galaxy Run

```
python "Foreground Masking\optimise_sep_spike_gate_parameters.py" --max-images 20 --initial-points 16 --max-iter 64
```

Resume an Interrupted Run

```
python "Foreground Masking\optimise_sep_spike_gate_parameters.py" --resume-output-dir "D:\Dropbox\Public Documents\UCLAN\MSc Research\Remove foreground objects\sep spike optimisation\YYYYMMDD_HHMMSS" --max-images 20 --initial-points 16 --max-iter 64
```

- Do not resume into a directory while the original process is still running.
- Terminal progress includes timestamped trial start/end lines and per-galaxy coverage/masking lines by default.
- Use --no-progress-galaxies to suppress per-galaxy lines.
- Use --prepare-only to inspect the selected spike-positive galaxies without running Optuna.

Issues and Caveats

- SEP is likely faster than MTObjects, but its detection threshold and deblend choices can be more sensitive to galaxy residual structure.
- Spike Gate is profile evidence, not ground-truth segmentation. Visual review remains necessary before accepting final parameters.
- The optimiser currently tunes global SEP parameters. A parameter set that covers spikes may still remove unrelated objects, which is partly intended but must be checked for science impact.
- Control galaxies can be included with --no-require-spikes if over-masking needs a stronger safeguard.

Systematic documentation review - 17 August 2026

Review classification: Reviewed - SEP Spike Gate reference remains valid

Updates made: No Toy Objects objective or batch-layout claim required correction. A dated scope note was added.

Current interpretation: This document remains specific to SEP Spike Gate optimisation and does not supersede the Toy Objects four-fold results.

Authoritative combined outcome: SEP and MTObjects Masking Comparison and Recommendations 20260817.docx. Full optimisation history and final science-image SEP fold records: Foreground Masking Optimisation Results.xlsx.